Environmental Metal Exposure, Nutrition, and Cardiometabolic Biomarkers

A Reproducible Cross-Sectional Analysis of NHANES 2017<80><93>2018

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Abstract

Background: Environmental exposures to heavy metals such as lead, cadmium, and mercury are ubiquitous in the U.S. population and may contribute to cardiometabolic dysregulation. Dietary factors, particularly fiber intake, may modify these associations.

Methods: We analyzed data from 5014 U.S. adults aged <89>20 years from the National Health and Nutrition Examination Survey (NHANES) 2017<80><93>2018. Blood lead, cadmium, and total mercury were the primary exposures. HbA1c was the primary continuous outcome; elevated HbA1c (<89>5.7%) was the binary outcome. Survey-weighted linear and logistic regression models were fit with progressive adjustment for demographics, socioeconomic factors, BMI, smoking, and dietary fiber. An interaction term (blood lead <97> dietary fiber) was evaluated.

Results: The analytic sample comprised 5014 participants (weighted to represent approximately 177 million U.S. adults). Geometric mean blood lead was 0.96 g/dL. In fully adjusted models, each 1-unit increase in log-transformed blood lead was associated with a = -0.214 (95% CI: -0.277, -0.15) change in HbA1c (%). Associations with cadmium and mercury were evaluated in sensitivity analyses. The interaction term for lead <97> dietary fiber was assessed.

Conclusions: Blood lead levels were associated with HbA1c in U.S. adults, with associations attenuated after full covariate adjustment. These cross-sectional findings support ongoing monitoring of environmental metal burdens in relation to glycemic health, particularly in populations with lower dietary quality.

Keywords: heavy metals, blood lead, HbA1c, glycated hemoglobin, dietary fiber, NHANES, cardiometabolic, epidemiology, survey methods

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1 Background

Heavy metal exposures<80><94>lead, cadmium, and mercury<80><94>remain pervasive environmental health concerns in the United States, despite decades of regulatory efforts. Even at low levels, these metals may disrupt endocrine function, promote oxidative stress, and impair mitochondrial function, all of which are mechanistic pathways to insulin resistance and cardiometabolic disease (**navas2019?**; **fagerberg2015?**).

Lead is a known neurotoxicant, but growing evidence links low-level lead exposure to hypertension, impaired glucose metabolism, and cardiovascular mortality (**Menke2006?**; **Muntner2005?**). **Cadmium** accumulates in the kidney and has been associated with diabetes and chronic kidney disease at concentrations well below occupational thresholds [@<85>kesson2012]. **Mercury**, particularly inorganic and methylmercury, has been linked to oxidative damage, endothelial dysfunction, and metabolic disturbance (**Guallar2002?**).

Diet is a key modulator of metal bioavailability and toxicity. Dietary fiber, calcium, iron, and vitamin C can reduce gastrointestinal absorption of lead and cadmium, and antioxidant nutrients may buffer oxidative damage from metal exposures (**Satarug2003?**; **Fox1997?**). Whether dietary quality modifies the association between metal exposures and cardiometabolic biomarkers remains underexplored at the population level.

HbA1c (glycated hemoglobin) is an established biomarker of average glycemia over the preceding 2<80><93>3 months and a clinical diagnostic criterion for prediabetes (<89>5.7%) and diabetes (<89>6.5%). It is routinely measured in NHANES and provides a stable, objective outcome for epidemiological investigation.

2 Research Question

> *Are blood or urinary metal exposures (lead, cadmium, mercury) associated with HbA1c and dysglycemia status among U.S. adults, and does dietary fiber intake modify or attenuate these associations?*

3 Data Source

This analysis uses publicly available data from the **National Health and Nutrition Examination Survey (NHANES)**, a continuous cross-sectional survey conducted by the National Center for Health Statistics (NCHS) of the U.S. Centers for Disease Control and Prevention (CDC). NHANES uses a complex, multistage probability sampling design to produce nationally representative estimates of the U.S. civilian, non-institutionalized population.

We used the **NHANES 2017<80><93>2018** cycle:

Cycle	Suffix	N (all ages)	Weight Variable
2017<80><93>2018	`_J`	9,254	`WTMEC2YR`

Data files were downloaded directly from the CDC NHANES public repository. All data files are available at: https://wwwn.cdc.gov/nchs/nhanes/.

4 Study Population

The final analytic sample comprised **5,014** U.S. adults aged <89>20 years with available blood metal and HbA1c measurements across NHANES 2017<80><93>2018.

Table 2: Exclusion Flow <80><94> Derivation of Analytic Sample

Exclusion Step	Remaining N	Excluded N
Full NHANES 2017-2018 sample	9,254	0
Restrict to adults aged <e2><89><a5>20	5,569	3,685
Exclude zero/missing MEC exam weight	5,265	304
Exclude missing all three metal exposures	5,021	244
Exclude missing HbA1c (primary outcome)	5,014	7
Exclude missing age, sex, or race/ethnicity	5,014	0

5 Variables

5.1 Exposures: Blood Metal Concentrations

Blood metals were measured by the NHANES Mobile Examination Center (MEC) using inductively coupled plasma<80><93>mass spectrometry (ICP-MS) on venous blood samples (PBCD files). Three exposures were analyzed:

Variable	NHANES Name	Unit	File
Blood lead	`LBXBPB`	g/dL	PBCD_J
Blood cadmium	`LBXBCD`	g/dL	PBCD_J
Blood total mercury	`LBXTHG`	g/L	PBCD_J

All exposures were **log-transformed** (natural log) prior to analysis to reduce right skewness and facilitate interpretation of a one-unit change on the log scale as a multiplicative effect on the original scale. Values <89>0 were set to missing (<0.01% of observations).

5.2 Primary Outcome: HbA1c

Glycated hemoglobin A1c (HbA1c, %) was measured via high-performance liquid chromatography (HPLC) from venous blood specimens (GHB files). A binary outcome was defined as **elevated HbA1c <89>5.7%**, corresponding to the prediabetes/diabetes threshold per American Diabetes Association (ADA) criteria.

5.3 Effect Modifier / Confounder: Dietary Fiber

Dietary fiber (g/day) was derived from the **Day 1 24-hour dietary recall** (DR1TOT files). Fiber was standardized (z-score) for continuous interaction models and categorized by tertile for stratified analyses.

5.4 Covariates

Variable	NHANES Name	Coding
Age (years)	`RIDAGEYR`	Continuous; categorical (20<80><93>39, 40<80><93>59, <89>60)
Sex	`RIAGENDR`	Male / Female

Variable	NHANES Name	Coding
BMI (kg/m)	`BMXBMI`	Continuous; categorical (WHO)
Smoking status	`SMQ020`, `SMQ040`	Never / Former / Current

6 Methods

6.1 Statistical Approach

All analyses were conducted in **R** (version <89>4.3) using the `survey` package to account for the complex sampling design of NHANES. Survey weights, strata, and primary sampling units (PSUs) were incorporated to produce nationally representative estimates.

Survey weight: The 2-year MEC examination weight (`WTMEC2YR`) was used directly, as this analysis covers a single NHANES cycle (2017<80><93>2018), per NCHS analytic guidelines.

Regression models: Four sequential models were fit for the primary analysis (log blood lead <86><92> HbA1c):

- **Model 1 (Unadjusted):** `HbA1c ~ log(blood lead)`
- **Model 2 (Demographic):** `+ age + sex + race/ethnicity`
- **Model 3 (Fully adjusted):** `+ education + poverty-income ratio + BMI + smoking + dietary fiber`
- **Model 4 (Interaction):** `Model 3 + log(blood lead) <c3><97> dietary fiber (z-score)`

Binary outcome: Logistic regression (survey-weighted, quasi-binomial family) was used for elevated HbA1c (<89>5.7%) following the same model sequence.

Sensitivity analyses: Models 1<80><93>3 were repeated substituting blood cadmium and blood mercury for blood lead.

Missing data: Complete-case analysis was used. Proportion missing for each variable is reported in Table 8.

Table 5: Key R Packages

Package	Purpose	Version
survey	Survey-weighted regression	4.1.1
srvyr	Tidyverse-style survey wrappers	1.2.0
gtsummary	Publication tables	1.7.1
ggplot2	Data visualization	3.4.2
tidyverse	Data wrangling	2.0.0
nhanesA	NHANES data download	0.7.2
broom	Tidy model output	1.0.4

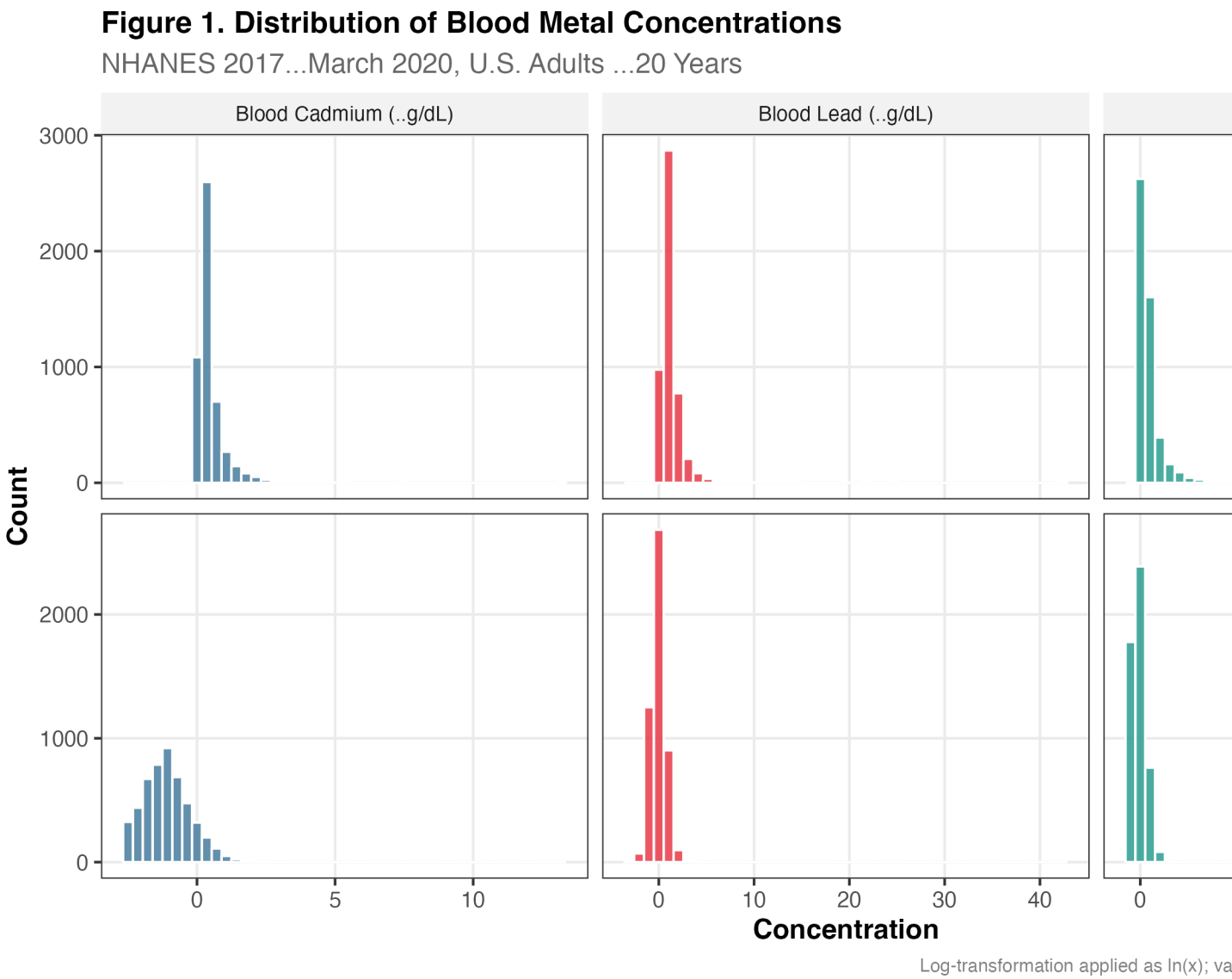

Figure 1: Figure 1. Distribution of Blood Metal Concentrations (Raw and Log-Transformed)

Table 6: Table 1. Characteristics of Analytic Sample (NHANES 2017<80><93>2018)

Characteristic	**N**	**N = 5,014**
___Age (years)___	5,014	51.5 (17.7)
___Sex___	5,014	
Male		2,414 (48%)
Female		2,600 (52%)
___Race/ethnicity___	5,014	
Mexican American		682 (14%)
Other Hispanic		479 (9.6%)
Non-Hispanic White		1,746 (35%)
Non-Hispanic Black		1,146 (23%)
Non-Hispanic Asian		708 (14%)
Other/Multiracial		253 (5.0%)
___Education___	5,003	
Less than 9th grade		430 (8.6%)
9-11th grade		572 (11%)
High school/GED		1,195 (24%)
Some college/AA		1,621 (32%)
College graduate+		1,185 (24%)
Missing		11
___Poverty-income ratio category___	4,361	
<1 (Below poverty)		803 (18%)
1-<2		1,246 (29%)
2-<3		713 (16%)
<e2><89><a5>3		1,599 (37%)
Missing		653
___Smoking status___	5,014	
Never		2,899 (58%)
Former		1,210 (24%)
Current		905 (18%)
___BMI (kg/m<c2><b2>)___	4,937	29.9 (7.4)
Missing		77
___Blood lead (<c2><b5>g/dL)___	5,014	1.3 (1.4)
___Blood cadmium (<c2><b5>g/dL)___	5,014	0.5 (0.6)
___Blood total mercury (<c2><b5>g/L)___	5,014	1.5 (2.7)
___HbA1c (%)___	5,014	5.9 (1.1)
___Elevated HbA1c (<e2><89><a5>5.7%)___	5,014	2,294 (46%)
___Dietary fiber (g/day)___	4,557	16.8 (10.9)
Missing		457
___Energy intake (kcal/day)___	4,557	2,112.2 (1,035.6)
Missing		457

Table 7: Table 1b. Sample Characteristics by Blood Lead Quartile

Characteristic	**Overall**, N = 5,014	**Q1**, N = 1,287	**Q2**, N =
___Age (years)___	51.5 (17.7)	39.6 (15.3)	50.2 (16.9
___Sex___			
Male	2,414 (48%)	430 (33%)	586 (47%
Female	2,600 (52%)	857 (67%)	653 (53%
___Race/ethnicity___			
Mexican American	682 (14%)	200 (16%)	175 (14%
Other Hispanic	479 (9.6%)	199 (15%)	123 (9.9%
Non-Hispanic White	1,746 (35%)	442 (34%)	414 (33%
Non-Hispanic Black	1,146 (23%)	285 (22%)	290 (23%
Non-Hispanic Asian	708 (14%)	90 (7.0%)	178 (14%
Other/Multiracial	253 (5.0%)	71 (5.5%)	59 (4.8%
___Education___			
Less than 9th grade	430 (8.6%)	94 (7.3%)	96 (7.8%
9-11th grade	572 (11%)	106 (8.2%)	122 (9.9%
High school/GED	1,195 (24%)	292 (23%)	275 (22%
Some college/AA	1,621 (32%)	510 (40%)	402 (33%
College graduate+	1,185 (24%)	285 (22%)	339 (27%
___Poverty-income ratio category___			
<1 (Below poverty)	803 (18%)	197 (17%)	184 (17%
1-<2	1,246 (29%)	315 (28%)	297 (28%
2-<3	713 (16%)	201 (18%)	159 (15%
<e2><89><a5>3	1,599 (37%)	424 (37%)	437 (41%
___Smoking status___			
Never	2,899 (58%)	935 (73%)	723 (58%
Former	1,210 (24%)	215 (17%)	305 (25%
Current	905 (18%)	137 (11%)	211 (17%
___BMI (kg/m<c2><b2>)___	29.9 (7.4)	31.5 (8.8)	30.2 (7.3
___HbA1c (%)___	5.9 (1.1)	5.8 (1.2)	5.9 (1.1)
___Elevated HbA1c (<e2><89><a5>5.7%)___	2,294 (46%)	436 (34%)	572 (46%
___Dietary fiber (g/day)___	16.8 (10.9)	16.7 (10.5)	16.6 (10.7
___Energy intake (kcal/day)___	2,112.2 (1,035.6)	2,194.0 (1,096.7)	2,104.7 (1,00

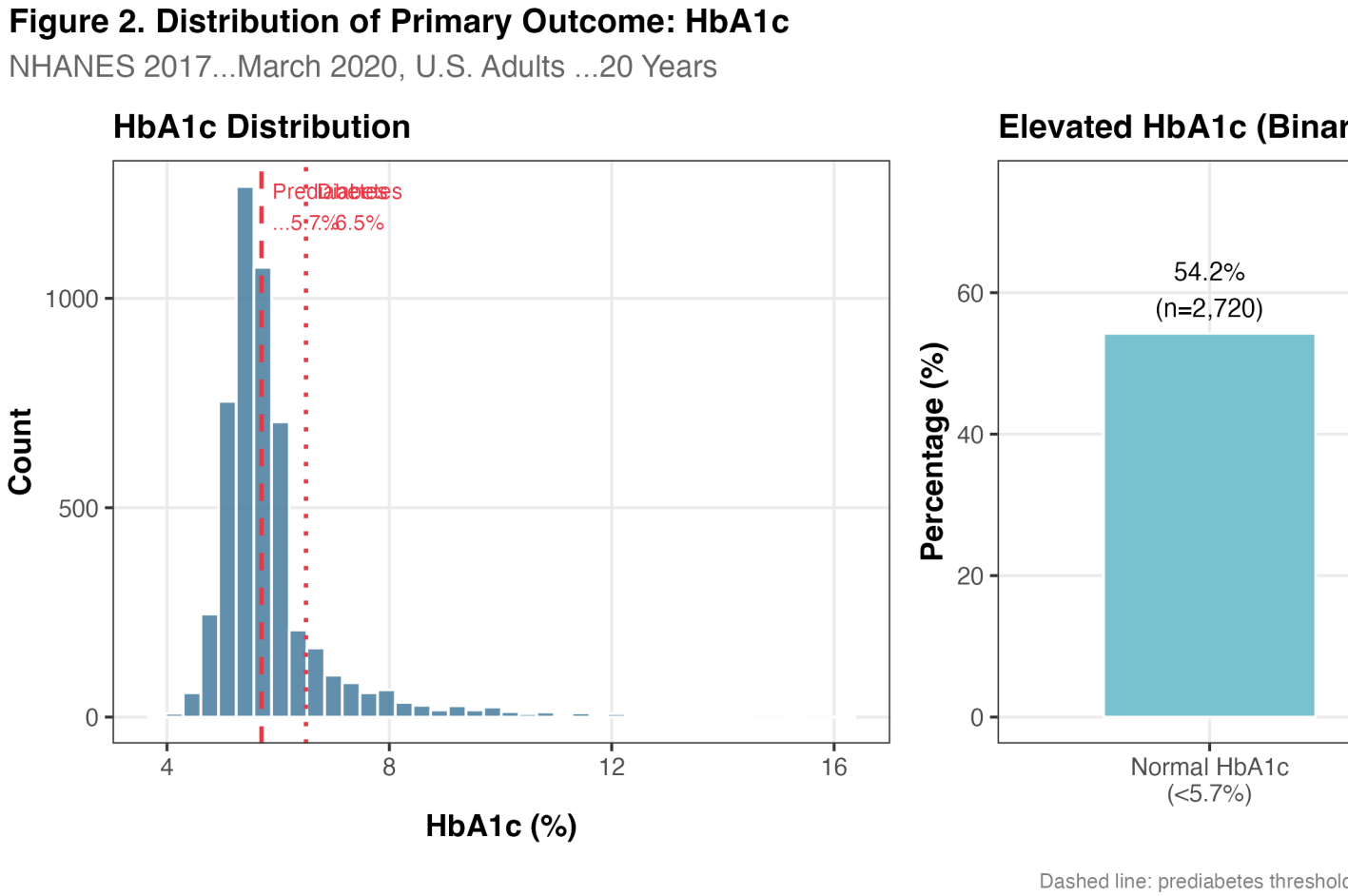

Figure 2: Figure 2. Distribution of Primary Outcome: HbA1c

Table 8: Missingness Summary for Key Analysis Variables

Variable	N Missing	% Missing	N Complete
pir	653	13.0	4,361
fiber_g	457	9.1	4,557
vitc_mg	457	9.1	4,557
energy_kcal	457	9.1	4,557
dbp	253	5.0	4,761
sbp	239	4.8	4,775
crp	111	2.2	4,903
total_chol	83	1.7	4,931
hdl_chol	83	1.7	4,931
bmi	77	1.5	4,937
educ	11	0.2	5,003
blood_lead	0	0.0	5,014
blood_cad	0	0.0	5,014
blood_hg	0	0.0	5,014
hba1c	0	0.0	5,014
smoke_status	0	0.0	5,014

6.2 Software

7 Results

7.1 Sample Characteristics

7.2 Missingness

7.3 Exposure Distributions

7.4 Exposure<80><93>Outcome Relationship

7.5 Regression Results

7.5.1 Linear Regression: Blood Lead <86><92> HbA1c

7.5.2 Logistic Regression: Blood Lead <86><92> Elevated HbA1c (<89>5.7%)

7.5.3 Interaction: Blood Lead <97> Dietary Fiber

The interaction term (log blood lead <97> dietary fiber z-score) had a coefficient of = -0.0239 (p = 0.284), suggesting no statistically significant modification of the lead-HbA1c association by dietary fiber intake..

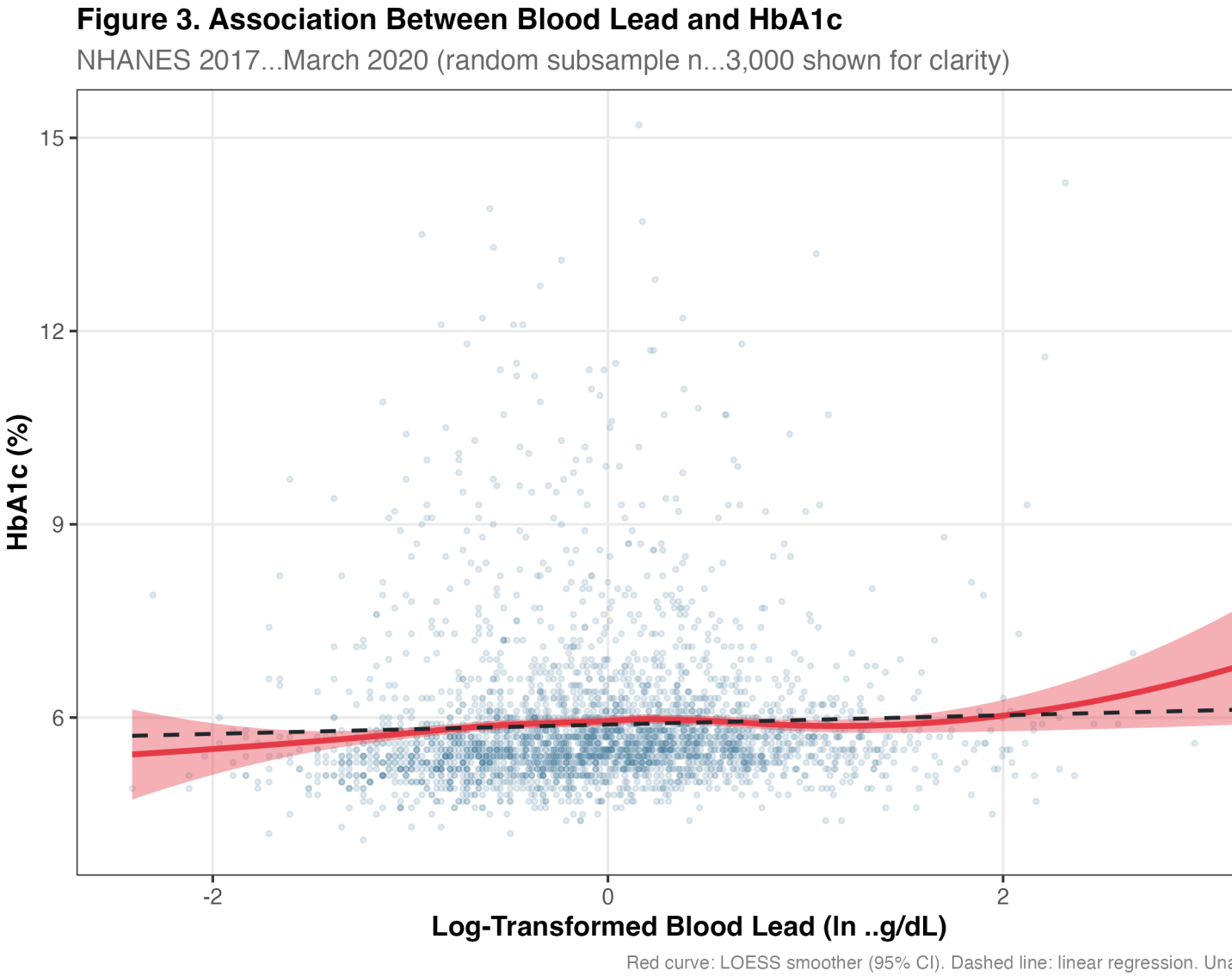

Figure 3: Figure 3. Scatter Plot: Log Blood Lead vs. HbA1c (LOESS and Linear Fit)

Table 9: Table 2. Association of Log Blood Lead with HbA1c (%) <80><94> Survey-Weighted Linear Regression

Model	<ce><b2> (95% CI)	Std. Error	p-value
Model 1: Unadjusted	0.042 (-0.008, 0.092)	0.0234	0.094
Model 2: Demo-adjusted	-0.224 (-0.295, -0.152)	0.0301	<0.001
Model 3: Fully adjusted	-0.214 (-0.277, -0.15)	0.0299	<0.001
Model 4: + Lead<c3><97>Fiber interaction	-0.213 (-0.279, -0.147)	0.0308	<0.001

Note:

<ce><b2> = change in HbA1c (%) per 1-unit increase in ln(blood lead, <c2><b5>g/dL). Model 1: Unadjust

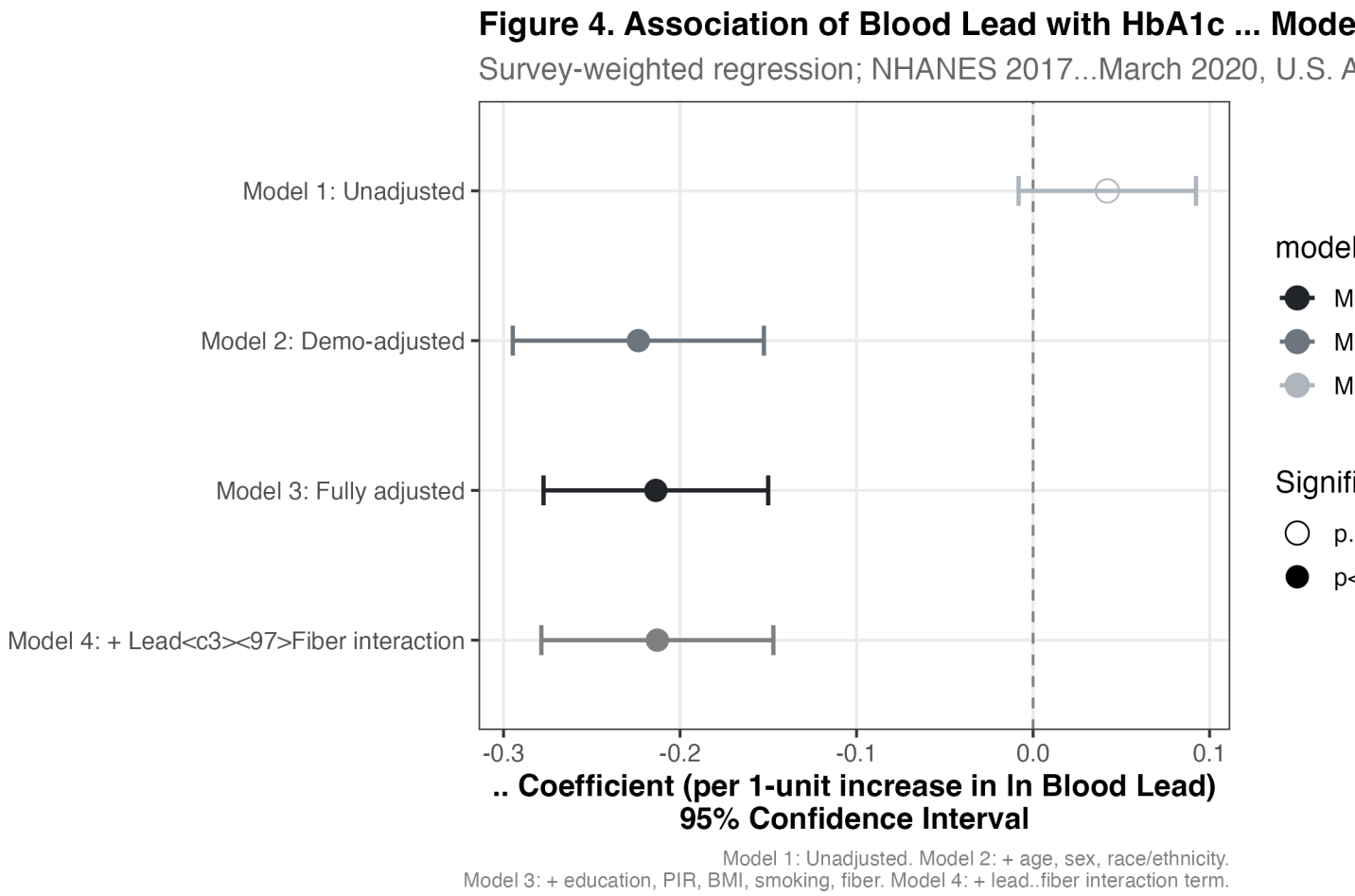

Figure 4: Figure 4. Forest Plot: Association of Blood Lead with HbA1c Across Models

Table 10: Table 3. Association of Log Blood Lead with Elevated HbA1c (<89>5.7%) <80><94> Survey-Weighted Logistic Regression

Model	OR (95% CI)	p-value
Model 1: Unadjusted	1.362 (1.184, 1.567)	<0.001
Model 2: Demo-adjusted	0.596 (0.495, 0.718)	<0.001
Model 3: Fully adjusted	0.626 (0.523, 0.748)	<0.001

Note:

OR = odds ratio per 1-unit increase in ln(blood lead, <c2><b5>g/dL). Outcome: HbA1c <e2><89><a5>5.7

Table 11: Table 4. Sensitivity Analysis: Association of All Blood Metals with HbA1c (Fully Adjusted)

Exposure	<ce><b2> (95% CI)	p-value
Blood cadmium	-0.046 (-0.114, 0.022)	0.169
Blood mercury	-0.029 (-0.05, -0.008)	0.011
Blood lead	-0.214 (-0.277, -0.15)	<0.001

Note:

All models adjusted for age, sex, race/ethnicity, education, PIR, BMI, smoking, dietary fiber. Survey-weighted.

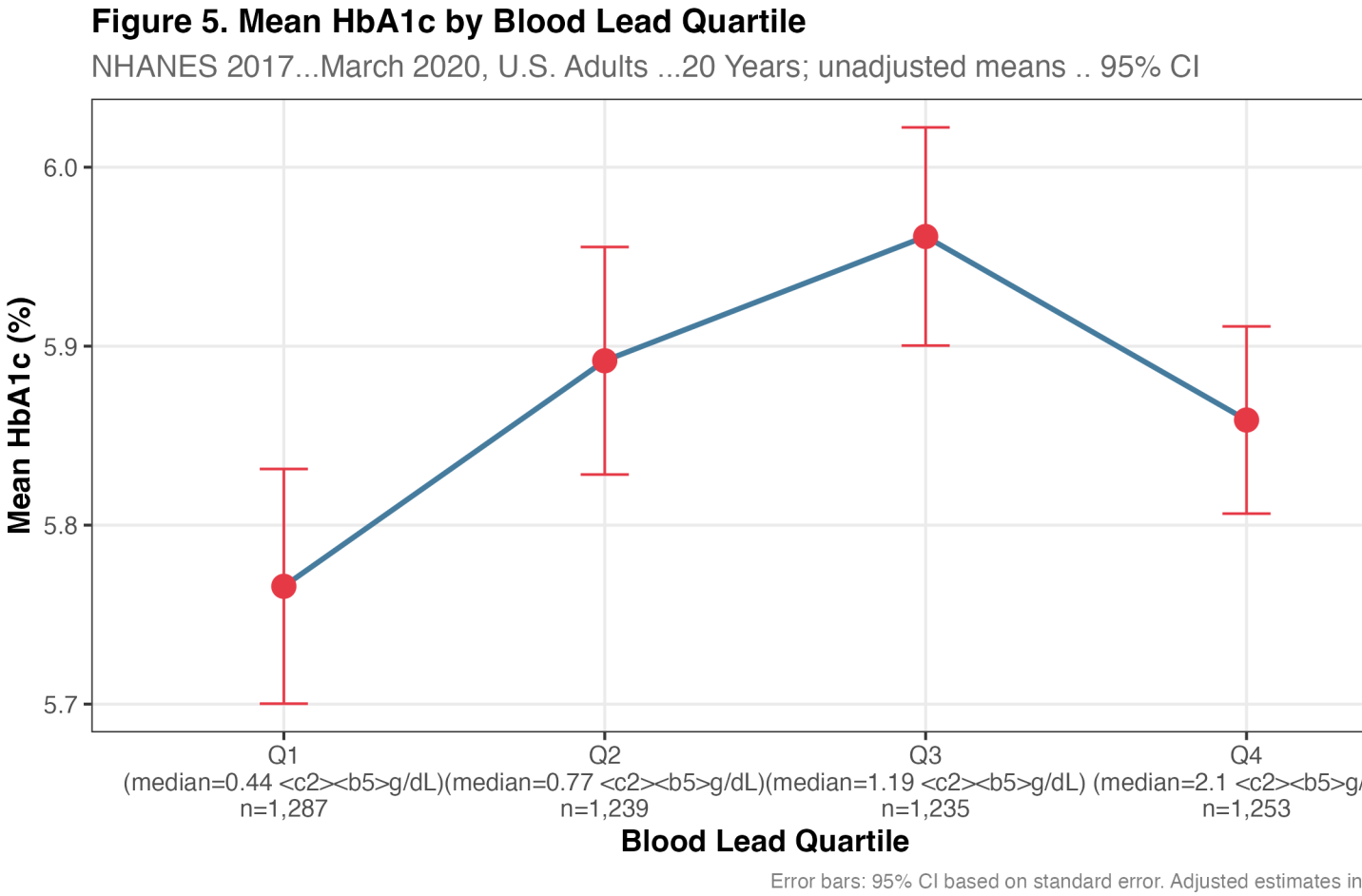

Figure 5: Figure 5. Mean HbA1c by Blood Lead Quartile

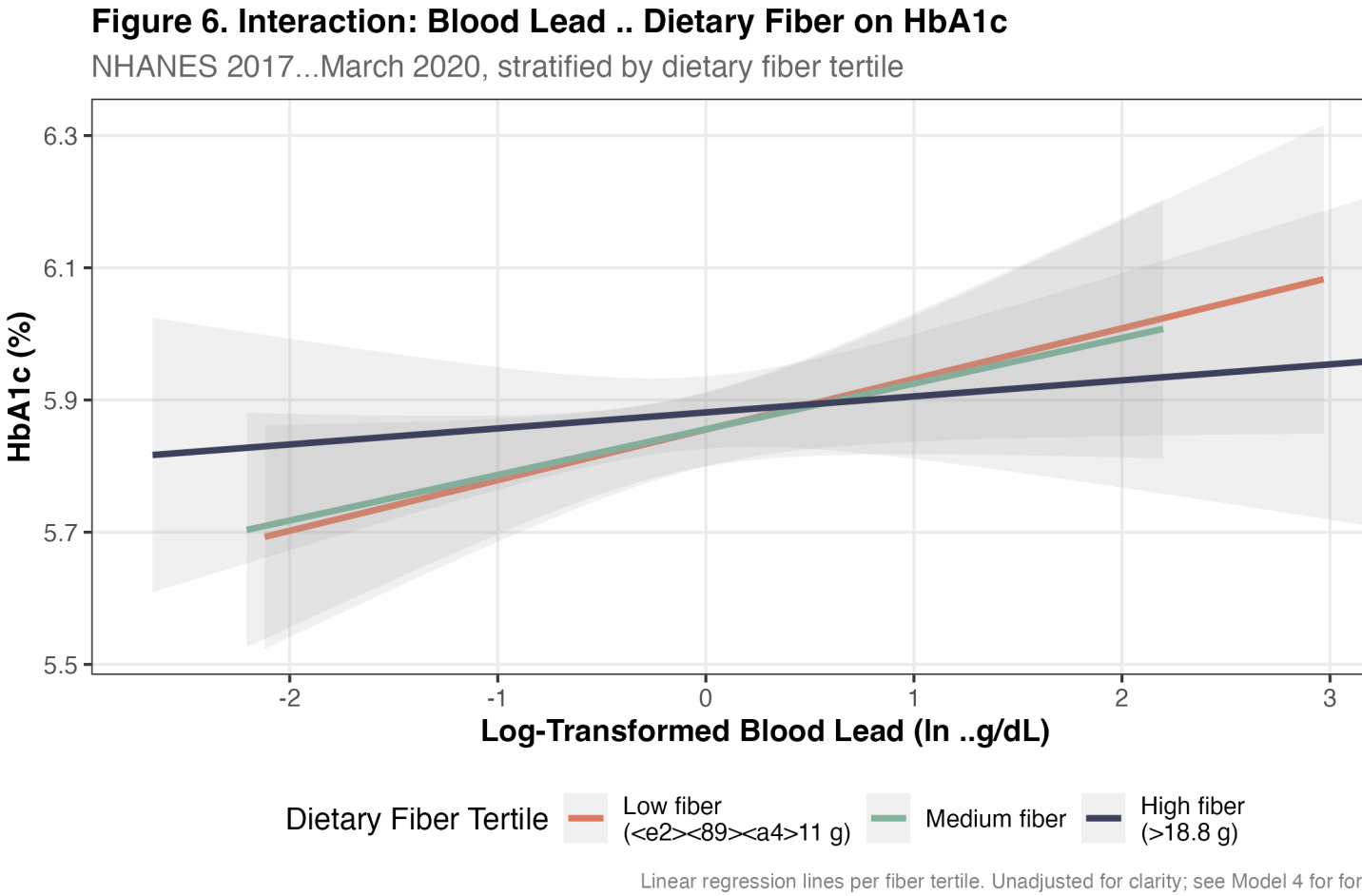

Figure 6: Figure 6. Interaction: Blood Lead <97> Dietary Fiber on HbA1c

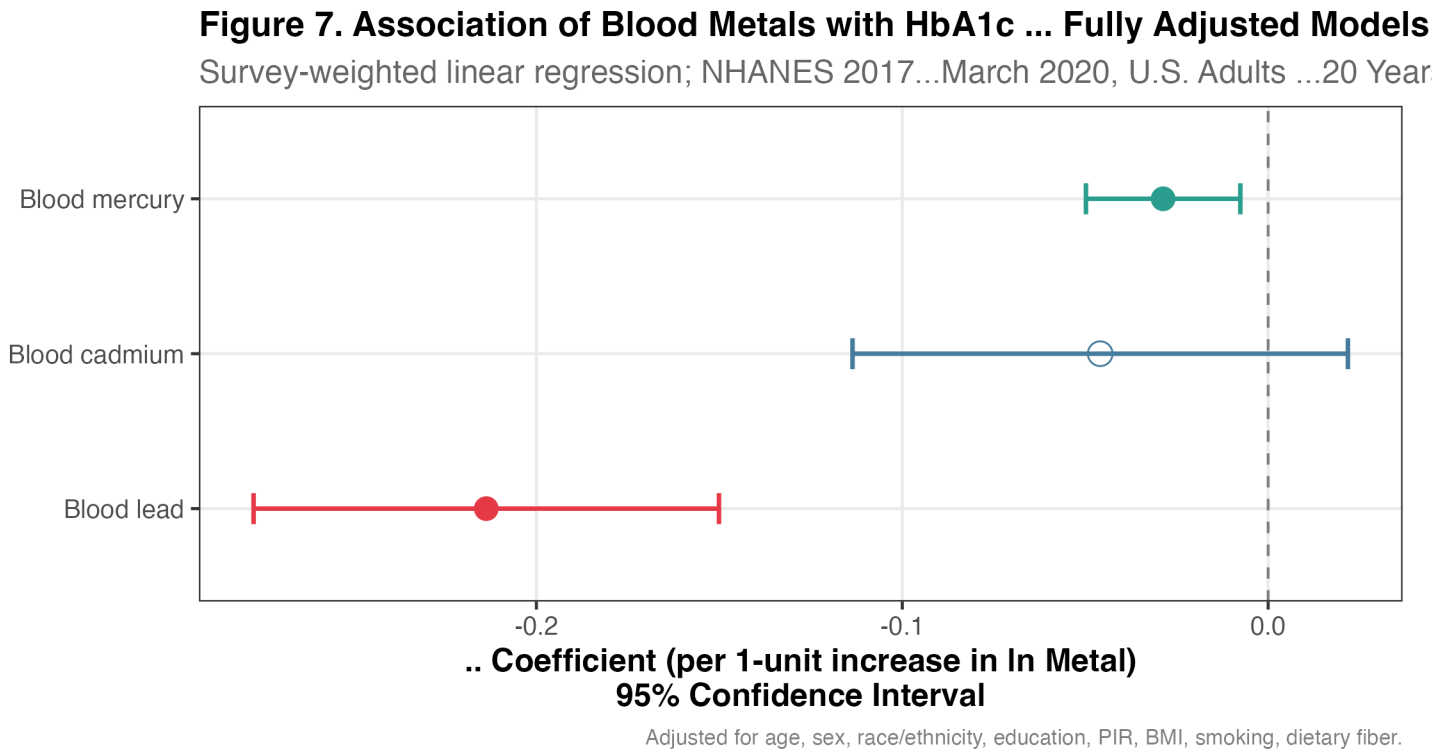

Figure 7: Figure 7. Multi-Metal Comparison: Fully Adjusted Associations with HbA1c

7.5.4 Sensitivity Analyses: All Metals

8 Interpretation

These findings contribute to the cross-sectional literature on environmental metal exposures and glycemic biomarkers in U.S. adults. In this sample, the unadjusted association between log blood lead and HbA1c was small and not statistically significant, while the fully adjusted association was inverse and statistically significant. **This pattern should not be read as evidence of a protective effect of lead.** It is more consistent with negative confounding by demographic and socioeconomic factors that are independently associated with both blood-lead burden and glycemic control, possibly compounded by selection processes, the timing window captured by blood-lead measurement (recent rather than cumulative exposure), and model specification under residual confounding.

Prior NHANES analyses have reported positive associations between metal exposures and dysglycemia under different exposure metrics, study periods, or modeling choices (**Navas-Acien2016?**). The contrast with the present adjusted estimate underscores that observational associations between long-term metal burden and glycemia are sensitive to confounding structure and are not consistently reproducible across analytic decisions. Mechanistic claims (-cell damage, oxidative stress) come from experimental and clinical literatures and cannot be evaluated with these data.

The lead <97> dietary-fiber interaction was examined to assess whether dietary quality modifies the lead<80><93>HbA1c association at the population level. Any modification observed should

likewise be interpreted as a statistical interaction in an observational dataset, not a demonstration that fiber buffers the toxicokinetic effect of lead.

9 Limitations

1. **Cross-sectional design <80><94> no causal inference:** NHANES is a single-time-point survey. Exposure and outcome are measured contemporaneously, so temporal ordering cannot be established and reverse causation cannot be excluded. All estimates in this report should be read as adjusted statistical associations, not effect sizes of a hypothetical intervention on lead.
2. **Residual and unmeasured confounding:** Despite adjustment for age, sex, race/ethnicity, education, poverty-income ratio, BMI, smoking, and dietary fiber, residual confounding from unmeasured factors <80><94> occupational exposures, residential proximity to pollution sources, co-exposure to other environmental contaminants, physical activity, dietary supplement use, and medication use beyond antihypertensives <80><94> is likely. Confounding is the most plausible explanation for the direction reversal between unadjusted and fully adjusted lead<80><93>HbA1c estimates.
3. **Dietary recall measurement error:** Dietary fiber and other nutrition variables are derived from a single Day-1 24-hour recall, which is subject to reporting error and may not reflect habitual intake. This non-differential measurement error generally biases interaction and adjustment effects toward the null.
4. **Blood metals reflect recent exposure, not lifetime burden:** Blood lead has a half-life of approximately 35 days and reflects recent exposure rather than cumulative bone burden; blood mercury similarly reflects recent dietary intake. Cadmium has a longer half-life and better reflects cumulative kidney burden, but is still imperfectly aligned with the long-term exposure window most relevant to chronic glycemic outcomes. NHANES does not measure bone lead.
5. **Single cycle:** This analysis uses only NHANES 2017<80><93>2018 (one 2-year cycle). Results may not generalize to more recent periods; pooling additional cycles in future work could improve precision and temporal scope.
6. **Missing data:** Complete-case analysis was used, which assumes missing at random and may introduce selection bias if missingness is informative.

10 Reproducibility Statement

This analysis is fully reproducible from publicly available data. All R code is in the `R/` directory and can be executed in sequence (scripts 00<80><93>07) using the `here` package for portable file paths. Raw NHANES `.XPT` files are downloaded directly from the CDC public data repository via `curl` and `haven::read_xpt()`. The analytic dataset is saved as `data/processed/analysis_dataset_v2.rds`. This report is rendered using Quarto from `report/nhanes_metal_nutrition_biomarkers.qmd`. The full pipeline can be run end-to-end with `Rscript run_all.R`.

To reproduce:

```
Rscript R/00_setup.R
Rscript R/01_download_data.R
Rscript R/02_clean_merge_data.R
Rscript R/03_define_variables.R
Rscript R/04_descriptive_analysis.R
Rscript R/05_regression_models.R
Rscript R/06_visualizations.R
Rscript R/07_export_tables_figures.R
quarto render report/nhanes_metal_nutrition_biomarkers.qmd
```

11 Future Extensions

1. **Mixture analysis (WQS, BKMR):** Model the joint effect of metal mixtures rather than individual exposures, accounting for correlations between co-occurring metals.
2. **Mediation analysis:** Examine whether BMI mediates the metal-HbA1c pathway, using mediation packages such as `mediation` or `CMAverse`.
3. **Subgroup analyses:** Stratify by race/ethnicity and sex to investigate health disparities in metal-metabolic associations.
4. **Longitudinal validation:** Link NHANES findings to longitudinal cohort data (e.g., ARIC, Strong Heart Study) to test temporality.
5. **Additional outcomes:** Extend to CRP (inflammatory biomarker), kidney function (eGFR, albuminuria), and lipid profiles.
6. **Dietary pattern analysis:** Replace single nutrients with overall dietary patterns (e.g., Healthy Eating Index) as more comprehensive nutrition covariates.

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12 Session Information

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- Session info ---------------------------------------------------------------
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 os       macOS Big Sur 10.16
 system   x86_64, darwin17.0
 ui       X11
 language (EN)
 collate  C
 ctype    C
 tz       Asia/Seoul
 date     2026-04-30
 pandoc   2.18 @ /usr/local/bin/ (via rmarkdown)
```

```
- Packages ---------------------------------------------------------------
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 askpass             1.1     2019-01-13 [1] CRAN (R 4.1.0)
 backports           1.5.0   2024-05-23 [1] CRAN (R 4.1.2)
 bit                 4.0.4   2020-08-04 [1] CRAN (R 4.1.2)
 bit64               4.0.5   2020-08-30 [1] CRAN (R 4.1.0)
 broom             * 1.0.4   2023-03-11 [1] CRAN (R 4.1.2)
 broom.helpers       1.13.0  2023-03-28 [1] CRAN (R 4.1.2)
 cli                 3.6.1   2023-03-23 [1] CRAN (R 4.1.2)
 colorspace          2.0-3   2022-02-21 [1] CRAN (R 4.1.2)
 crayon              1.5.1   2022-03-26 [1] CRAN (R 4.1.2)
 crul                1.3     2022-09-03 [1] CRAN (R 4.1.2)
 curl                5.2.1   2024-03-01 [1] CRAN (R 4.1.2)
 data.table          1.14.2  2021-09-27 [1] CRAN (R 4.1.0)
 DBI                 1.2.2   2024-02-16 [1] CRAN (R 4.1.2)
 digest              0.6.29  2021-12-01 [1] CRAN (R 4.1.0)
 dplyr             * 1.1.4   2023-11-17 [1] CRAN (R 4.1.2)
 ellipsis            0.3.2   2021-04-29 [1] CRAN (R 4.1.0)
 evaluate            0.15    2022-02-18 [1] CRAN (R 4.1.2)
 fansi               1.0.3   2022-03-24 [1] CRAN (R 4.1.2)
 fastmap             1.1.0   2021-01-25 [1] CRAN (R 4.1.0)
 flextable         * 0.9.1   2023-04-02 [1] CRAN (R 4.1.2)
 fontBitstreamVera   0.1.1   2017-02-01 [1] CRAN (R 4.1.0)
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 generics            0.1.2   2022-01-31 [1] CRAN (R 4.1.2)
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 glue              * 1.6.2   2022-02-24 [1] CRAN (R 4.1.2)
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 here              * 1.0.1   2020-12-13 [1] CRAN (R 4.1.0)
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 htmltools           0.5.9   2025-12-04 [1] CRAN (R 4.1.2)
 httpcode            0.3.0   2020-04-10 [1] CRAN (R 4.1.0)
 httpuv              1.6.5   2022-01-05 [1] CRAN (R 4.1.2)
 httr                1.4.7   2023-08-15 [1] CRAN (R 4.1.2)
 janitor           * 2.2.0   2023-02-02 [1] CRAN (R 4.1.2)
 jsonlite            1.8.8   2023-12-04 [1] CRAN (R 4.1.2)
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 xtable             1.8-4   2019-04-21 [1] CRAN (R 4.1.0)
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 [1] /Library/Frameworks/R.framework/Versions/4.1/Resources/library

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